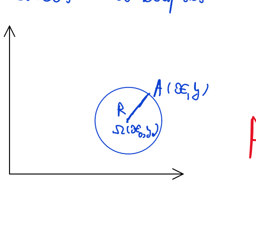
Seminar 11

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Circles and ellipses



$\mathcal{C}(\Omega,R): (x-x_0)^2+(y-y_0)^2=R^2$ ↳ Implicit form; $(x-x_0)^2+(y-y_0)^2$ = dist(A,Ω)

Parametric form: $\mathcal{C}(\Omega,R)$ ↳ Parametric form: $\begin{cases} x=x_0+R\cos t \\ y=y_0+R\sin t \end{cases}$; $t\in[0,2\pi)$

Circles

1. Find the equation of the circle:

a) of diameter $[A,B]$, with $A(1,2)$ and $B(-3,-1)$,

b) with center $I(2,-3)$ and radius $R=7$,

c) with center $I(-1,2)$ and passing through $A(2,6)$,

d) centered at the origin and tangent to $\ell: 3x-4y+20=0$,

e) passing through $A(3,1)$ and $B(-1,3)$ and having the center on the line $\ell: 3x-y-2=0$,

f) passing through $A(1,1)$, $B(1,-1)$ and $C(2,0)$,

g) tangent to both $\ell_1: 2x+y-5=0$ and $\ell_2: 2x+y+15=0$ if one tangency point is $M(3,-1)$.

a) $[A,B]$ diameter $\Rightarrow \begin{cases} x_\Omega=\frac{x_A+x_B}{2}=-1 \\ y_\Omega=\frac{y_A+y_B}{2}=\frac{1}{2} \end{cases}$

$AB=\sqrt{(x_A-x_B)^2+(y_A-y_B)^2}=\sqrt{4^2+3^2}=5 \Rightarrow R=\frac{AB}{2}=\frac{5}{2}$

$(x-x_\Omega)^2+(y-y_\Omega)^2=R^2$

$(x+1)^2+(y-\frac{1}{2})^2=\frac{25}{4} \Leftrightarrow x^2+2x+1+y^2-y+\frac{1}{4}=\frac{25}{4} \Leftrightarrow x^2+2x+y^2-y=11$

b) $(x-2)^2+(y-3)^2=7^2$

$x^2-4x+4+y^2-6y+9=49$

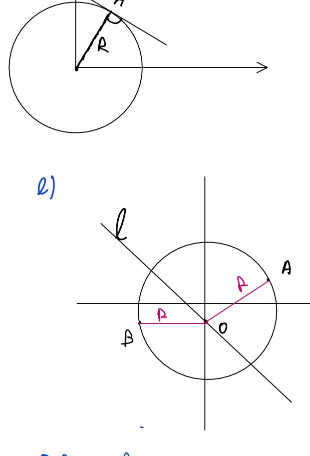
$x^2-4x+y^2-6y=36$

c) $R=iA=\sqrt{3^2+4^2}=5$

$(x+1)^2+(y-2)^2=25$

$x^2+2x+y^2-4y=20$

d)

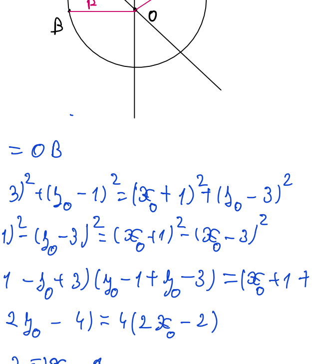


$d(0,\ell)=R=\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$

$d(0,\ell)=\frac{20}{5}=4$

$x^2+y^2=16$

e)



$OA=OB$

$(x_0-3)^2+(y_0-1)^2=(x_0+1)^2+(y_0-3)^2$

$(y_0-1)^2-(y_0-3)^2=(x_0+1)^2-(x_0-3)^2$

$(y_0-1-y_0+3)(y_0-1+y_0-3)=(x_0+1+x_0-3)(x_0+1-x_0+3)$

$2(2y_0-4)=4(2x_0-2)$

$y_0-2=2x_0-2$

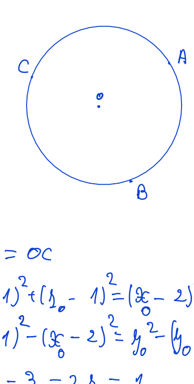
$\begin{cases} y_0=2x_0 \\ 3x_0-y_0-2=0 \end{cases} \Leftrightarrow \begin{cases} x_0=2 \\ y_0=4 \end{cases}$

$R=\sqrt{(2-3)^2+(4-1)^2}=\sqrt{10}$

$(x-2)^2+(y-4)^2=10$

$x^2-4x+y^2-8y=-10$

f)



$OA=OC$

$(x_0-1)^2+(y_0-1)^2=(x_0-2)^2+y_0^2$

$(x_0-1)^2-(x_0-2)^2=y_0^2-(y_0-1)^2$

$2x_0-3=2y_0-1$

$x_0=y_0+1$

$OB=OC$

$(x_0-1)^2+(y_0+1)^2=(x_0-2)^2+y_0^2$

$2x_0-3=-(2y_0+1)$

$2x_0-3=-2y_0-1$

$x_0=-y_0+1$

$y_0+1=-y_0+1$

$\begin{cases} y_0=0 \\ x_0=1 \end{cases}$

g) $\ell_1: 2x+y-5=0$

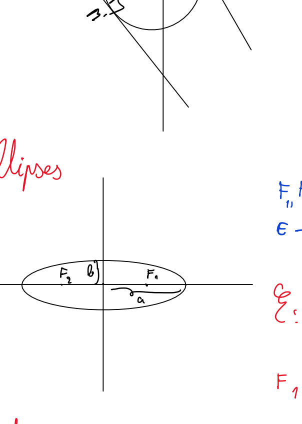
$\ell_2: 2x+y+15=0$

$M(3,-1)\in\ell_1$

$\ell_2: \underbrace{2x+y-5}_{\ell_1}+20 \Rightarrow \ell_1 \parallel \ell_2 \Rightarrow MM'$ diameter

$MM'=20$

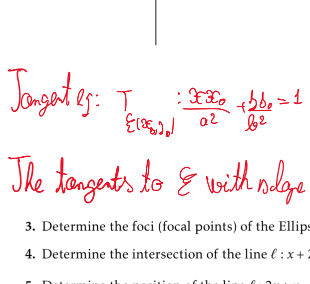
$\Rightarrow \Omega M=10=R$



$\Rightarrow \begin{cases} 2x_{M'}+y_{M'}=-15 \\ (x_{M'}-3)^2+(y_{M'}+1)^2=400 \end{cases}$

⋮

Ellipses



F_1, F_2 distinct points (foci of the ellipse)

$\mathcal{E}$ – locus of points P in the plane s.t. $PF_1+PF_2=2a$

$\mathcal{E}: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a\geq b$

$F_1(-c,0)$, $F_2(c,0)$, $c=\sqrt{a^2-b^2}$

Tangent eq.: $T_{\mathcal{E}(x_0,y_0)}: \frac{x x_0}{a^2}+\frac{y y_0}{b^2}=1$ $\quad e=\frac{c}{a}$ eccentricity of $\mathcal{E}$

The tangents to $\mathcal{E}$ with slope m: $y=mx\pm\sqrt{m^2a^2+b^2}$

3. Determine the foci (focal points) of the Ellipse $9x^2+25y^2-225=0$

4. Determine the intersection of the line $\ell: x+2y-7=0$ and the ellipse $\mathcal{E}: x^2+3y^2-25=0$.

5. Determine the position of the line $\ell: 2x+y-10=0$ relative to the ellipse $\mathcal{E}: \frac{x^2}{9}+\frac{y^2}{4}-1=0$.

6. Determine an equation of a line which is orthogonal to $\ell: 2x-2y-13=0$ and tangent to the ellipse $\mathcal{E}: x^2+4y^2-20=0$.

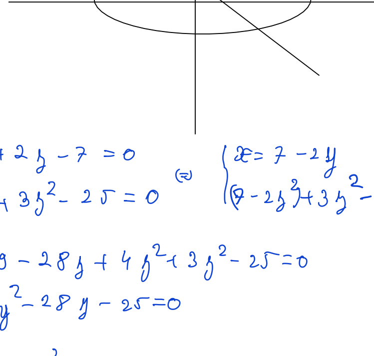
3) $9x^2+25y^2-225=0$

$9x^2+25y^2=225 \quad |:225$

$\frac{x^2}{5^2}+\frac{y^2}{3^2}=1 \Rightarrow a=5, b=3$

$c=\sqrt{5^2-3^2}=4 \Rightarrow F_1(-4,0), F_2(4,0)$

4)



$\begin{cases} x+2y-7=0 \\ x^2+3y^2-25=0 \end{cases} \Leftrightarrow \begin{cases} x=7-2y \\ (7-2y)^2+3y^2-25=0 \end{cases}$

$49-28y+4y^2+3y^2-25=0$

$7y^2-28y-25=0$

$\Delta=28^2+700$

$\Delta=1468$

$28\cdot28$: 208, 56, 768 $\quad 1468 \xrightarrow{2} 734 \xrightarrow{2} 367$

$y_1=\frac{28+2\sqrt{367}}{14}=2+\frac{\sqrt{367}}{7}$

$y_2=2-\frac{\sqrt{367}}{7}$

5) $\begin{cases} 2x+y-10=0 \Rightarrow 2x=10-y \\ (2x)^2+(3y)^2=36 \end{cases}$

$(10-y)^2+(3y)^2=36$

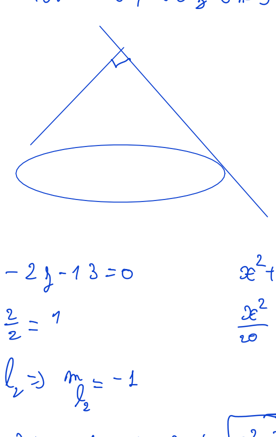
$100-20y+y^2+9y^2=36$

$100-20y+10y^2=36$

$(\sqrt{10}y)^2-2\cdot\sqrt{10}y\sqrt{10}+10+90=36$

$(\sqrt{10}y-\sqrt{10})^2=-54 \Rightarrow \nexists y\in\mathbb{R} \Rightarrow$ the line does not intersect the ellipse

6)



$\ell_1: 2x-2y-13=0$ $\qquad x^2+4y^2-20=0$

$m_{\ell_1}=\frac{2}{2}=1$ $\qquad \frac{x^2}{20}+\frac{y^2}{5}=0$

$\ell_1\perp\ell_2 \Rightarrow m_{\ell_2}=-1$

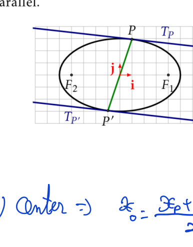
$T_{\ell_2}(x,y)$: $y=m\cdot x\pm\sqrt{a^2m^2+b^2}$

$y=-x\pm\sqrt{20+5}$

$y=-x+5$

$y=-x-5$

7. A *diameter* of an ellipse is the line segment determined by the intersection points of the ellipse with a line passing through the center of the ellipse. Show that the tangent lines to an ellipse at the endpoints of a diameter are parallel.



Consider $O(0,0)$ center $\Rightarrow x_0=\frac{x_P+x_{P'}}{2} \Rightarrow x_P=-x_{P'}$

$y_0=\frac{y_P+y_{P'}}{2} \Rightarrow y_P=-y_{P'}$

$y_P=m_1x_P\pm\sqrt{m_1^2a^2+b^2}$

$y_{P'}=m_2x_{P'}\pm\sqrt{m_2^2a^2+b^2}$

$-y_P=-m_2x_P\pm\sqrt{m_2^2a^2+b^2}$

$y_P=m_2x_P\mp\sqrt{m_2^2a^2+b^2}$

$\Rightarrow m_1=m_2 \Rightarrow$ the lines are parallel

8. Consider the family of ellipses $\mathcal{E}_a: \frac{x^2}{a^2}+\frac{y^2}{16}=1$. For what value $a\in\mathbb{R}$ is $\mathcal{E}_a$ tangent to the line $\ell: x-y+5=0$?

9. Consider the family of lines $\ell_c: \sqrt{5}x-y+c=0$. For what values $c\in\mathbb{R}$ is ℓ_c tangent to the ellipse $\mathcal{E}: x^2+\frac{y^2}{4}=1$?

8) $\mathcal{E}_a: \frac{x^2}{a^2}+\frac{y^2}{16}=1$

$\begin{cases} 16x^2+a^2y^2=16a^2 \\ x=y-5 \end{cases}$

$16(y-5)^2+a^2y^2=16a^2$

$16(y^2-10y+25)+a^2y^2-16a^2=0$

$(16+a^2)y^2-160y+400-16a^2=0$

$(16+a^2)y^2-160y-16(16+a^2)+656=0$

Let $b=16+a^2$, $b\geq16$

$by^2-160y-16b+656=0$

$\Delta=160^2+4b(16b-656)\geq0$

$160^2+4\cdot16b(b-41)\geq0$

$160\cdot10+4b(b-41)\geq0$

$400+b^2-41b\geq0$

$b^2-41b+400\geq0$

$\Delta_2=41^2-1600=81$

$b_1=\frac{41+9}{2}=25 \Rightarrow 16+a^2\geq25$ $\quad a\in\mathbb{R}\setminus(-3,3)$

$b_2=\frac{41-9}{2}=16 \Rightarrow 16+a^2\leq16$ (False)

9) $\ell_c=\sqrt{5}x-y+c=0 \Rightarrow y=\sqrt{5}x+c \Rightarrow y^2=5x^2+2\sqrt{5}cx+c^2$

$x^2+\frac{y^2}{4}=1 \quad |\cdot4$

$4x^2+y^2=4$

$4x^2+5x^2+2c\sqrt{5}x+c^2=4$

$9x^2+2c\sqrt{5}x+c^2-4=0$

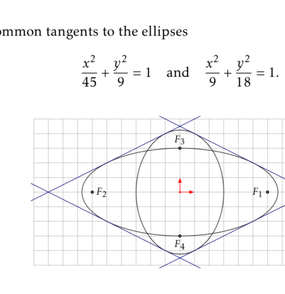
$\Delta=20c^2-36(c^2-4)\geq0$

$-16c^2+4\cdot4\cdot9\geq0$

$c^2\leq9 \Rightarrow c\in[-3,3]$

10. Determine the common tangents to the ellipses

$$\frac{x^2}{45}+\frac{y^2}{9}=1 \quad\text{and}\quad \frac{x^2}{9}+\frac{y^2}{18}=1.$$



10) $\begin{cases} y=mx+\sqrt{45m^2+9} \\ y=mx+\sqrt{9m^2+18} \end{cases} \Rightarrow 45m^2+9=9m^2+18$, $m=\pm\frac{1}{2}$

$y=\pm\frac{1}{2}x+\sqrt{5\cdot\frac{1}{4}\ldots}$

$y=\pm\frac{1}{2}x\pm\frac{9}{2} \Leftrightarrow x=\pm2y\pm9$

11. Consider the ellipse $\mathcal{E}: \frac{x^2}{4}+y^2-1=0$ with focal points F_1 and F_2. Determine the points M, situated on the ellipse, for which

a) the angle $\measuredangle F_1MF_2$ is right;

b) the angle $\measuredangle F_1MF_2$ is θ;

c) the angle $\measuredangle F_1MF_2$ is maximal.

$\frac{x^2}{4}+y^2-1=0$

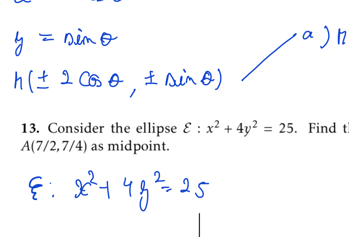
$x=2\cos\theta$

$y=\sin\theta$

$M(\pm2\cos\theta,\pm\sin\theta)$ — a) $M(0,1)$ and $M(0,-1)$

13. Consider the ellipse $\mathcal{E}: x^2+4y^2=25$. Find the chords on the ellipse which have the point $A(7/2, 7/4)$ as midpoint.

$\mathcal{E}: x^2+4y^2=25$



Let (x_1,y_1) and (x_2,y_2) be the points on the ellipse

$x_A=\frac{x_1+x_2}{2} \Rightarrow x_1+x_2=7$

$y_A=\frac{y_1+y_2}{2} \Rightarrow y_1+y_2=\frac{7}{2} \Rightarrow 2(y_1+y_2)=7$

$x_1^2+\frac{y_1^2}{4}=25 \quad |\cdot4$

$\begin{cases} 4x_1^2+y_1^2=100 \\ 4x_2^2+y_2^2=100 \\ x_2=7-x_1 \\ y_2=\frac{7-2y_1}{2} \end{cases}$

$4(x_1^2-x_2^2)=(y_2^2-y_1^2)$

$4(x_1-x_2)(x_1+x_2)=(y_2-y_1)(y_1+y_2)$

$28(x_1-x_2)=(y_2-y_1)\frac{7}{2} \quad |\cdot\frac{2}{7}$

$8(x_1-x_2)=y_2-y_1$

$8(x_1+x_2-2x_2)=-y_2-y_1+2y_2$

$8(7-2x_2)=-\frac{7}{2}+2y_2 \quad |\cdot2$

$16(7-2x_2)=-7+4y_2$

$17\cdot7=2x_2+4y_2$

$2x+4y-119=0$

14. Consider the ellipse $\mathcal{E}: \frac{x^2}{25}+\frac{y^2}{9}=1$. Determine the geometric locus of the midpoints of the chords on the ellipse which are parallel to the line $\ell: x+2y=1$.

$\frac{x^2}{25}+\frac{y^2}{9}=1 \Rightarrow a=5$, $b=3$

Locus: $\mathcal{E}=\{P \mid PF_1+PF_2=2a\}$

$F_1(-c,0)$ $F_2(c,0)$, $c=\sqrt{a^2-b^2}=4$

$P(x_p,y_p)$

$PF_1=\sqrt{(x_p-4)^2+y_p^2}$

$PF_2=\sqrt{(x_p+4)^2+y_p^2}$

$\begin{cases} \frac{x^2}{25}+\frac{y^2}{9}=1 \\ y=-\frac{1}{2}x+C \end{cases} \Rightarrow \frac{x^2}{25}+\frac{\left(-\frac{1}{2}x+C\right)^2}{9}=1$

$\frac{x^2}{25}+\frac{(2C-x)^2}{4\cdot3^2}=1$

$\frac{x^2}{25}+\frac{4C^2-4Cx+x^2}{4\cdot3^2}-1=0$

$\frac{x^2}{25}+\frac{C^2}{3^2}-\frac{xC}{3^2}+\frac{1}{36}x^2-1=0$

$\frac{61x^2}{25\cdot36}-\frac{C}{9}\cdot x+\frac{C^2-9}{9}=0$

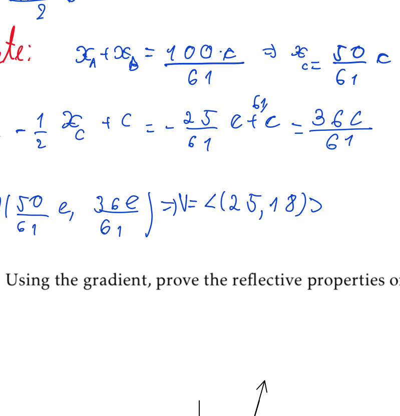
$x_C=\frac{x_A+x_B}{2}$

Viete: $x_A+x_B=\frac{100\cdot C}{61} \Rightarrow x_C=\frac{50}{61}C$

$y_C=-\frac{1}{2}x_C+C=-\frac{25}{61}C+C=\frac{36C}{61}$

$\Rightarrow M\left(\frac{50}{61}C, \frac{36C}{61}\right) \Rightarrow V=\langle(25,18)\rangle$

15. Using the gradient, prove the reflective properties of an ellipse.



$F_1(-c,0)$

$F_2(c,0)$

$f(x,y)=\frac{x^2}{a^2}+\frac{y^2}{b^2}$

$\nabla f=\left(\frac{2x}{a^2},\frac{2y}{b^2}\right)$

$\nabla f(x_M,y_M)=\left(\frac{2x_M}{a^2},\frac{y_M}{b^2}\right)$

Param eq: $\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} x_M \\ y_M \end{bmatrix}+t\cdot\begin{bmatrix} \frac{2x_M}{a^2} \\ \frac{2y_M}{b^2} \end{bmatrix}$

$N\in O_x \Rightarrow y_N=0$

$y_M+t\cdot\frac{2y_M}{b^2}=0$

$y_M\left(1+\frac{2t}{b^2}\right)=0 \Rightarrow 1+\frac{2t}{b^2}=0$

$2t+b^2=0$

$t=-\frac{b^2}{2}$

$x_N=x_M-\frac{b^2}{2}\cdot\frac{2x_M}{a^2}$

$x_N=x_M-\frac{b^2}{a^2}x_M$

$x_N=\frac{a^2-b^2}{a^2}$

$x_N=\frac{c^2}{a^2}x_M$

$MF_1=\sqrt{(-c-x_M)^2+y_M^2}=\sqrt{(c+x_M)^2+y_M^2}$

$MF_2=\sqrt{(x_M-c)^2+y_M^2}=\sqrt{(x_M-c)^2+y_M^2}$

$F_1N=\sqrt{\left(\frac{c^2}{a^2}x_M+c\right)^2}$

$F_2N=\sqrt{\left(\frac{c^2}{a^2}x_M-c\right)^2}$

$\frac{F_1N}{F_2N}=\frac{\frac{c^2}{a^2}x_M+c}{\frac{c^2}{a^2}x_M-c}=1+\frac{2c}{\frac{c^2}{a^2}x_M-c}=\sqrt{1+\frac{2}{\frac{c}{a^2}x_M-1}}$

$\frac{MF_1}{MF_2}=\frac{(c+x_M)^2+y_M^2}{(x_M-c)^2+y_M^2}=\frac{(c+x_M)^2-x_M^2+a^2b^2}{(x_M-c)^2-x_M^2+c^2b^2}=c^2-a^2b^2$

$\frac{x_M^2}{a^2}+\frac{y_M^2}{b^2}=1 \quad |\cdot a^2b^2$

$y_M^2=a^2b^2-x_M^2$